

INTRODUCTION

Agriculture remains the cornerstone of India's economy, supporting nearly half of the population and providing employment, nutrition, and rural development. The Kharif season, spanning from June to October, is particularly vital due to its reliance on the south-west monsoon. Approximately 55% of India's net sown area is rainfed, making the timely and equitable distribution of monsoon rains crucial for agricultural success.

In 2025, the India Meteorological Department (IMD) forecasts an above-normal monsoon season, with cumulative rainfall expected at 105% of the Long Period Average (LPA) of 87 cm. The monsoon's onset is anticipated to be slightly delayed in southern India but should reach Maharashtra by May 27, facilitating timely sowing preparations. The IMD's forecast of an above-normal monsoon offers a promising outlook but still carries risks of intra-seasonal dry spells and localized flooding. The Indian government's initiatives, such as the "Panchayat Mausam Seva," aim to deliver hyper-local weather forecasts at the panchayat level, ensuring that farmers receive accurate and timely information.

The Kharif season witnesses the cultivation of a diverse array of crops across India's agro-climatic zones, including rice, maize, cotton, soybean, groundnut, bajra, jowar, tur, urad, and moong. Despite advancements in agricultural technology, the sector remains vulnerable to climatic fluctuations such as delayed rains, uneven distribution, and extreme weather events. These challenges can adversely affect crop yields, farm incomes, and rural livelihoods.

To mitigate these risks, it is imperative to provide farmers with location-specific, research-backed guidance. Agro-advisories play a pivotal role by offering timely, region-specific, science-based information on weather patterns, crop management, pest control, harvesting, and storage. These advisories integrate seasonal outlooks with crop calendars and state-wise sowing windows, enabling farmers to select appropriate crop varieties and sowing dates, optimize resource use, and prevent yield losses.

This advisory serves as a crucial tool for agricultural officers, Farmer Producer Organizations (FPOs), and NGOs involved in rural development. By bridging the information gap between research and practice, it empowers farmers with the knowledge needed to make informed decisions, thereby contributing to the vision of a self-reliant, prosperous, and inclusive India @ VIKSIT BHARAT 2047.

case of stormy weather conditions, while heaping of incompletely dried paddy sheaves, 25 kg of salt per acre of paddy sheaves may be sprinkled and heaped to avoid damage.

Maize

In normal conditions, rainfed crop should be sown between June 15 and July 15. In the Kharif season, under rainfed conditions, maize should be sown only after a substantial rainfall (more than 50 mm). Seed rate is 17.5-20 kg/ha for normal hybrids, 7.5-10 kg/ha for sweetcorn, 12.5 kg/ha for popcorn and 25 kg/ha for baby corn. Within two to three days after sowing the crop, spray Atrazine herbicide @ 2 kg/ha in light soils and 3 kg/ha in heavy soils mixed with 500 liters of water when there is sufficient moisture which helps in controlling broadleaved and some grassy weeds for about 30 DAS. To control broadleaved and grassy weeds in normal and zero tillage maize crop at 15-18 DAS or 4 leaf weed stage, Tembotrione 34.4% SC @ 287.5 ml/ha + Atrazine 50% WP @ 1000 g/ha or Topramezone 33.6% SC 75 ml + Atrazine 50% WP @ 1000 g/ha should be sprayed. If the problem of *Cyperus spp.* is more, spray Halosulfuron methyl @ 100 g/ha mixed with 500 L of water.

Fall army worm: Summer ploughing, collection & destruction of egg masses, installation of pheromone traps (*S. frugiperda*) @ 4/acre sowing of border crop with grain sorghum & inter crop with cowpea (few rows) spraying of Azadirachtin 10000 ppm @ 2ml/l (10 to 15 DAS), EPN or Bt spray @ 2 ml/l (15 to 21 DAS). First insecticide spray - Emamectin Benzoate 5SG @ 0.4g/l (or) Spinosad 480SC @ 0.5 ml/l (21-28 DAS). Metarhizium anisopliae spray (1x107) @ 2 ml/l (30 -35DAS). Second Insecticide spray - Flubendiamide 480SC @ 0.3 ml/l (or) Chlorontriliniprole 18.5SC @ 0.3ml/l or Spinetoram 11.7 SC @ 0.3 ml/l (36 - 42 DAS). Poison Baiting - 45-65 DAS using Thiodicarb 75WP (250 g thiodicarb + 25kg rice bran + 5 kg jaggery / ha).

OILSEED CROPS

Groundnut

Farmers are advised to take up summer ploughing and also suggested to select high yielding varieties like TCGS 1694, K-Chitravathi, K-Lepakshi. The suitable time for sowing of rainfed groundnut is July to first week of August. Farmers are advised to adopt intercropping with redgram at 7:1/11:1/15:1 ratio as per their convenience and availability of sowing equipment. Use fertilizer cum seed drill instead of manual labour for sowing behind the country plough. Formation of conservation furrows every 3.6m interval at sowing/20-30 days after sowing will help in soil moisture conservation.

Farmers are also suggested to follow seed treatment with imidacloprid 600 FS @ 1ml per kg seed and followed by tebuconazole 2 DS @ 1g or mancozeb @ 3 g/kg or *Trichoderma viride* @ 10g/kg seed. Take up sowing of bajra/jowar (4-8 rows) as border crops all around the field to avoid transmission of vectors like thrips which spread PSND/PBND diseases. Spray KNO₃ @ 0.5% or 19:19:19 immediately after dry spell. Provide supplemental irrigation 20mm at flowering and peg penetration stage. Install sticky traps (10/acre) and pheromone traps (4/acre) in field to monitor

sucking and leaf eating pests. Intensity of infestation to spray imidacloprid 17.8 SL @ 0.3 ml/l of water for the control of sucking pests at ETL. Install bird perches (4/acre for collection and destruction of egg, larval populations by nesting birds). Spray quinalphos @ 2.0 ml and neem oil @ 5 ml/L of water at initial stages and at later stages. Spray emamectin benzoate 5% SG @ 0.4 g/l of water. Spray 2 g Chlorothalonil or 2 ml. Hexaconazole or 1 ml. Tebuconazole per liter of water for control of Tikka leaf spot and spray twice at an interval of 15 days. Spray Monocrotophos 1.6 ml or Fipronil 2 ml or Dimethoate 2 ml/L of water for control of peanut bud necrosis virus. To prevent this pest, treat the seeds with 2 ml imidacloprid 600 FS per kg of seed for the next crop.

Castor

Select high yielding varieties such as Haritha, Kiran, Jwala, Pragathi and hybrids i.e DCH. 117, PCH-111, GCH.8, ICH.66 and ICH 5. Optimum sowing time for castor is June 15th to July 31st. Seed rate of castor is 2-2.5 kg/acre and to fallow spacing 90 cm x 60 cm and suggested to take up thinning and remove excess population @ 10-15 DAS by retaining one seedling per hill and maintaining the optimum plant population. Farmers are advised to utilizing the available soil moisture and take up top dressing of urea @ 13 kg/acre at 30 to 35, 60 to 65 and for hybrids apply additional dose of urea @ 13 kg/acre at 90 to 95 days under sufficient soil moisture condition.

Sesamum

Farmers are advised to take up sowings during II FN of May and high yielding varieties like Sarada, Goutham, Varaha, Madhavi, and Gowri can be taken up. 2 kg of seeds per acre should be sown in rows mixed with sand or sawdust in equal proportions. Farmers are advised to do treat sesame seeds with 3 g of Carbendazim/Mancozeb per kg of seed. This will prevent leaf spot, root rot and stem rot. Sesame farmers should apply fertilizers that supply 16 kg of nitrogen, 8 kg of phosphorus, and 8 kg of potash per acre. Farmers are advised to spray Pendimethalin 30% @ 4-5 ml per litre of water to control weeds immediately after sowing in sesamum.

Sunflower

Farmers are advised to choose high-yielding hybrid sunflower such as NDSH-1012, LSFH-171, DRSH-1 and KBSH-44. Farmers are advised to take up intercrops such as groundnut and redgram (Sunflower: Groundnut-2:4 ratio) (Sunflower: redgram- 2:1 ratio). Farmers are advised to treat the seed with Iprodione 25% + Carbendazim 25%) or Fluxapyroxad FS @ 1.5 g/kg or Saaf (Carbendazim 12% + Mancozeb 63%) @ 2 g/kg for controlling alternaria leaf spot. In areas prone to necrosis virus in sunflower, treat seeds with Thiamethoxam (Cruiser) @ 4 g/kg or Imidacloprid (Gaucho) @ 5 g/kg. Farmers are advised to apply (26 kg urea) at sowing, 13 kg at 30 days during the bud stage, and 13 kg at 50 days during flowering. Apply the full dose of phosphorus (150 kg SSP) and potash (20 kg MOP) during the final ploughing. In sulphur-deficient soils, apply 10 kg sulphur per acre in gypsum form (55 kg/acre) to improve oil content and yield. If boron deficiency is noticed in sunflower, spray 2 g of borax per litre of water using 200 litre per acre.

Bajra

Use high yielding varieties like A.B.V.04, I.C.M.V. 221, I.C.M.V. 155, Raj 171 and PC 612. The seed rate of bajra @ 1.6 kg/acre and follow spacing of 45 cm x 15 cm. The seed should be soaked in 2% salt water solution for ten minutes and remove the inert material that floats on top of the solution. After drying, treat the seed with metalaxyl @ 6g/kg seed. Farmers are suggested to take up intercropping of Bajra: Redgram/ Groundnut in 2:1 ratio.

Finger Millet

Cultivate blast tolerant varieties like Vegavathi, CFMV 4, Indravathi, Suvarnamuki and Gosthani. Take up intercropping of ragi and redgram 8:2 ratio and Ragi: field bean 8:1 ratio.

Foxtail millet

Crops like jowar, bajra, foxtail millet and pulses viz., greengram, balckgram, horsegram, cowpea, field bean are used as contingency corps after July month due to delay in monsoon if at all. In chickpea areas, farmers are advised to take up foxtail millet as a preceding crop to chickpea. Where foxtail millet is grown during Kharif and chickpea during Rabi. Farmers are advised to cultivate high yielding varieties of foxtail millet like SIA 3156, renadu, mahanandi and garuda. Short duration Foxtail millet variety like SIA 3222 (Garuda) may be sown as preceding crop in rainfed black soils instead of keeping fallow.

Little millet

Farmers are advised to cultivate varieties of little millet like BL 6, DHLM 36-3 and OLM 203 for higher productivity.

FRUIT & VEGETABLE CROPS

Cashew

Prune during June-July and ploughing in bearing orchards.

Coconut

Cultivate turmeric, ginger, yam, colacasia, seasonal vegetables, pulses and ground nut in young gardens of below 5 years age. Plant cocoa, banana, pepper, pineapple, papaya, heliconia as intercrops in old and bearing gardens.

Oilpalm

Cultivate vegetables, dwarf banana varieties, flower crops, chilli, ginger, turmeric, pineapple as intercrops in young garden and cocoa and pepper in old and bearing gardens. Use of harvesting poles can reduce labour use. Timely harvesting of matured bunches to avoid damage due to kharif rains.

Mango

Installation of pheromone traps @ 12 per hectare to manage fruit fly. Mango orchards should be irrigated and rest should be allowed for 50 days after harvest. Basins should be dusted with Methyl Parathion dust. Prune crisscross and dried branches and left over fruit stalks during June-July, followed by spray 1% Bordeaux mixture. Fertilisers should be applied as per the leaf analysis i.e. trees more than 10 years age and in full bearing should be applied with N P and K @ 1 kg each (2.2 kg Urea, 1.6 kg

enrich the soil. Tractor operated FYM Pulverizer cum spreader is useful for FYM application in the field without any labour. It covers up to 10 to 12 acres per day. Cost of the implement is Rs. 76000/-.

DSR with Seed cum fertilizer drill increases 14% higher yields compared to traditional transplanting. Crops matured 7-10 days earlier. Farmers saved on 12 men labour and 25 women labour per hectare leading to a 17 % reduction in cost of cultivation. Tractor drawn Ananta groundnut planter (8 rows) is used for groundnut sowing. The intercrops of redgram or castor can also be sown with this implement. It covers 6 to 7 ha per day. For small and marginal farmers, use Handpush seed drill in light soils for Maize, Castor, Redgram, Cowpea, field bean and other bold seeded crops and in heavy soils on ridges sowing should be done for better crop stand establishment. Motor-Operated Groundnut Sheller is capable of shelling 25 to 250 kg of pods per hour. The distance between the seed drum and the concave can be adjusted according to pod size for efficient shelling. Powered by a 1 or 2 HP motor/engine depending on size. Use cono weeders for Drum seeder/transplanted rice in wet conditions. Power weeders help in upland rice systems or DSR & Early-stage weeding (15 to 30 DAS) is crucial for effective control. Self-propelled Weeder in paddy fields is operated by a 3 to 5 HP diesel engine, it can cover 0.15-0.25 acres per hour at low speed. & with a 5 to 6.5 HP engine, it can cover up to 1 acre per hour. Cost of cultivation is Rs.52,000 to Rs.56,000/ha.

Rain pipe irrigation system is effortless to install, reallocate and reduce the consumption of energy and water by more than 50 percent. These pipes are light weight, portable and flexible. Works on low water pressure. Expected life for 3 to 5 Years. Solar Fence system is used to control Wild boars a major problem in groundnut, paddy, maize and other crops. The solar fence cost varies from Rs. 15,000 to 25,000 by different companies like Pavani Solar systems, Nellore (Mobile numbers: 9848582332 & 9440171938) and Krupa Solar fence, Hyderabad. This unit protects 5 acres field. Use Acoustics (Sound emitting machinery) to repel the Birds, Wild boars, Monkeys etc., in crops like Groundnut, Sunflower, Maize, jowar etc. Use Bird scarer to repel the birds in the fields. The implement will create sound by rotating fan when natural air occurs. The cost is Rs.800/-. Straw Baling Machine is used to bundle paddy straw into either round or square/rectangular shapes for easy handling and transport. ANGRAU blade guntaka (Groundnut Digger) separates groundnut pods from the soil efficiently without wasting time. Built to dig up to 30 cm depth and can harvest up to 1 acre per hour.

Groundnut Digger, shaker cum Windower digs plants up to about 25 cm depth, shakes off the soil, and arranges the plants into windrows behind the machine. Cost: Rs.1,68,000/-. Groundnut Stripper separates pods from the plant by using a 1 to 3 HP motor. Requires two labour. Capacity: 100-125 kg per hour. Tractor operated pit digger for sugarcane planting will dig pits 250 to 300 holes per hour with a diameter of 90 cm. Cost is Rs.1,20,000. Cutting sugarcane sett with single bud by Sugarcane sett cutter. Capacity about 1500 setts per hour and Cost of the implement is Rs.5000/-. Damage is less when compared to manual cutting. Sugarcane detrasher can be used for detrashing. Easy for handling and detrashing, reduces the labour

requirement and Cost of the implement is Rs.1200/-. Cotton Picker mechanically plucks open cotton bolls from the plant without damaging the fiber. Used in large-scale operations.

Fruit Crops

Secateurs are used during June to August for pruning mango branches lesser than 1 inch thickness and for removing dry twigs. Cost of the falcon company implement is Rs. 450. Loppers are used during June to August for pruning mango branches and dry twigs less than 3 inches thickness. Cost of the falcon company implement is Rs. 1500. Mango harvester is used to harvest mangoes without damaging the fruits. Cost of the implement is Rs. 3150. Banana bunch cutter is used to harvest banana bunches. Coconut climbing cycles are used to easily climb coconut trees for harvesting coconuts and fronds and for pest and disease management. Cost of the implement is Rs. 3500/-. Coconut dehusker is used to remove the husk from coconut shells. Cost of the implement is Rs. 950. Oil palm harvester is used to harvest oil palm bunches, bring them down, and load them into a tractor.

Livestock

A milking machine is beneficial for farmers with more than 4-5 dairy cows as it extracts milk 3-4 times faster than manual milking and reduces physical labour. Farmers rearing more than 30-50 backyard poultry birds can use a vaccinator to administer the specified dose of vaccine in a very short time. When raising backyard poultry birds in large numbers, de-beaking with the help of a debeaker is a good management practice to reduce pecking. With the help of a duplex poultry shed, 10 - 15 chickens can be kept in a protective shelter providing with integrated feed and protection, and manage very easily at very low cost.

Drone Technology

Spraying of pesticides through Drones reduces time and cost of application and it is the Measure against labour scarcity. In Paddy, Maize, Jowar, Groundnut, Bengalgram, Chilli crops, this technology is already proved effective. An Acre can be sprayed with 10 litres capacity of Drone in 6-10 minutes.

ENTERPRISES

Adopt secondary agriculture through value addition to primary agriculture for enhancing farm income and sustainable rural livelihoods. Agripreneurship promotion through farm mechanization, crop diversification, processing, value addition, organic farming. Take up Agro based enterprises such as commercial horticulture, cultivation of high value crops, animal husbandry, fisheries, poultry, and Mushroom cultivation for doubling farm income. Setting up of Agro produce processing units like Mini dhal mills, Rice Mills and decorticating mills for promotion of agripreneurship among rural youth & women. Value addition of agriculture producing through Agro produce manufacturing units such as preparation of millet value added products, bakery products, ready to cook products, ready to eat products and nutraceuticals. Entrepreneurship by agro- input manufacturing units such as production of organic manures, fertilizers, agricultural inputs, vermin compost units, Bio fertilizers and agricultural implements. Establishments and

appearance of 5% damage symptoms. This should be followed by a spray of Emamectin Benzoate 5 SG at 0.4 g per litre of water, five days after sand application. Subsequently, a second spray using a combination formulation of Thiamethoxam 12.6% + Lambda-Cyhalothrin 9.5% @0.5 ml per litre of water should be administered 15 days after the first insecticidal application.

PULSE CROPS

Farmers are advised to select recommended varieties for cultivation of pulse crops are as follows: Pigeonpea/Arhar: Medium (150-160 days) UPAS-120, Bahar, Pusa9, Pusa-992, Asha and for long duration 170-180 days T-21, IPA-203, C-11. Birsa Arhar-1, Bahar, UPAS, IPA-203, Narendra Arhar-1, Narendra Arhar-2. Black Gram: Birsa Urad-1, WBU-09, Pant U-31. Green Gram: SML-668, IPM-2,3, HUM-16. Horse gram: Birsa Kulthi-1, Indira Kulthi-1, VLG-19. Farmers

are advised to select and sow pigeon pea in upland area where good water drainage facility is available. Use variety Narendra Arhar-1, Malviya Arhar (MAL-13) for drought tolerant area. Farmer are advice to harvest the crop when 80-85% pods turn brown but delay in harvesting may lead to pod shattering and yield loss. For effective management of Wilt (*Fusarium spp.*) farmers are advised to use resistant varieties (e.g., Bahar); practice crop rotation. For Podborer (*Helicoverpa armigera*) spray Spinosad 45 SC @0.3 ml/L or Emamectin Benzoate 5 SG @0.5 g/L at flowering and pod formation stages and Sterility Mosaic Virus can be controlled by removing infected plants; control whitefly/vector using Imidacloprid 17.8SL@0.5 ml/L.



Diagnostic visit in pulse field

OILSEED CROPS

Farmers are advice to select recommended varieties for cultivation of oil seed crops are as follows: Groundnut: K-1812, K-6, Dharni, Birsa Mungfali-3, Birsa Mungfali-4, Birsa Bold, Niger: Birsa Niger-1, Birsa Niger-3, Sesame: RT-351, Krishna, Kanke Safed, Shekhar, Soyabean: Birsa Safed Soyabean-2, JS-2098, Sunflower: DCS-107. Groundnut is a promising oilseed crop for the kharif season in the rainfed uplands and medium lands of Bihar Zones IIIA and IIIB. The optimal sowing period is between 15th June to 5th July, in line with monsoon onset. Avoid delay beyond mid-July to reduce risk of wilt and pod borer incidence. Seed rate: 12-15 kg/ha with Spacing: 60 cm (row to row) × 20 cm (plant to plant) is practice. Recommended varieties include TG-37A, JL-24, ICGV-91114, ICGS-44, and Girnar-2, which are early to medium maturing and resistant to foliar diseases. Groundnut seeds should be treated with Carbendazim or Mancozeb (2 g/kg) and inoculated with Rhizobium and PSB cultures to enhance nitrogen fixation and phosphorus uptake. Use a seed rate of 80-100 kg/ha and maintain 30 x 10 cm spacing for optimal plant population.

Weed management can be done and first intercultural operation 20-25 days after sowing and Second operation can be done 40-45 days after sowing. Apply Pendimethalin 1.0 kg a.i./ha as pre-emergence herbicide (within 2 days of sowing).



Goa's tropical monsoon climate, with high humidity and heavy rainfall, supports Kharif crops like paddy (the staple), pulses, groundnut, and vegetables. The region's lateritic and alluvial soils, ample rain, and moderate temperatures favor crop growth. To optimize yields and manage risks such as pest outbreaks and waterlogging, timely and location-specific practices are essential. This advisory provides key guidelines on land preparation, crop choice, pest and nutrient management, and sustainable farming for the Kharif season in Goa.

Prepare land for Kharif paddy and new orchards. Begin nursery sowing of Kharif rice
Crop-wise Advisory.

CEREAL CROPS

Paddy

Sowing of Kharif rice in nursery can be done. Procure good quality seeds, fertilizers and other inputs. As the Rainy season is approaching, all the farmers are advised to repair the bunds of rice fields and keep bunds weed free. While puddling the field, green leaf manures or biomass could be added and incorporated @5t/ha. When the intensity of rainfall is low, apply 1 kg urea per 100 m² to the rice nursery. Control the crabs in the rice field by collecting during night time.

FRUIT CROPS

Banana

Clear drainage; avoid waterlogging. Provide staking support to young plants

Mango

Harvest stage: Remove fallen fruits; orchard hygiene. Fruit rot risk: Hot water treat fruits (48°C, 1 hr). Anthracnose: Spray Carbendazim @2g/L on seedlings. New planting: Space 5×5m; fill pits with FYM, neem cake, rock phosphate



Gujarat provides region-specific, science-based recommendations to help farmers improve productivity and resilience during the monsoon season. With climate variability and resource constraints in mind, it promotes integrated crop management, timely pest-disease control, efficient water-nutrient use, and adoption of improved seed varieties. Tailored to the state's varied soils and rainfall patterns, this guidance aims to support sustainable farming and secure farmer livelihoods across Gujarat.

SOUTH SAURASHTRA

This region includes Amreli, Bhavnagar, Botad, Devbhumi-Dwarka, Gir-Somnath, Jamnagar, Junagadh, Morbi, Porbandar, Rajkot, and Surendranagar, SAU: Junagarh Agricultural University, Junagarh (Gujarat). They have an average rainfall of 400-1000 mm. Soils are shallow to medium black soils, often affected by salinity and alkalinity.

Groundnut

Groundnut cultivation begins with deep ploughing the field 2-3 times to achieve a fine tilth up to 25-30 cm depth. Incorporating 5-7 t/ha of well-decomposed farmyard manure (FYM) before sowing improves soil fertility. Sowing is generally done during June-July. Recommended semi-spreading varieties include GJG-22, GG-20, and Girnar-4, while erect types include GJG-32, GJG-17, and KDG-128. For optimal plant spacing, maintain 60 × 10 cm for semi-spreading types and 45 × 10 cm for erect ones. Seeds should be treated with Vitavax and *Rhizobium* at 375-500 g/ha to promote healthy germination and nitrogen fixation. Weed control involves pre-emergence application of Pendimethalin at 1.0 kg/ha, followed by two hand weedings at 20 and 40 days after sowing (DAS), or alternatively, post-emergence application of Quizalofop-P-ethyl at 50 g/ha. Timely irrigation is crucial, especially during flowering, pegging, and pod development stages in case of dry spells.

Nutrient management includes application of potassium at 50 kg/ha along with recommended nitrogen and phosphorus doses, and in saline soils, gypsum should be applied at 5 t/ha. To enhance crop nutrition, foliar spray of 0.2% boric acid or nano boron should be applied at 30, 45, and 60 DAS. For plant protection, collar and stem rot can be managed with a spray of Mancozeb or neem extract around 75 DAS. White grub infestation can be controlled through seed treatment with Chlorpyrifos or Imidacloprid, and soil application of *Metarhizium anisopliae* or Carbofuran. Sucking pests should be managed through seed treatment with Imidacloprid or foliar spray with Thiamethoxam, while defoliators can be controlled using Quinalphos, Chlorpyrifos, or Flubendiamide. Harvesting should be done at 120-125 DAS using a blade harrow, followed by threshing to separate the pods.



Groundnut Demo Plot

Soybean

Soybean grows best in well-drained loamy soils and should be avoided in saline or alkaline conditions. Deep ploughing and the application of farmyard manure (FYM) are essential for soil preparation. Recommended varieties include JS 335, GJS-3, and GS-4. Sowing is done in June-July with a spacing of 45×5 cm and a depth of 3-5 cm, using a seed rate of 70-80 kg/ha. Seeds should be treated with Carbendazim or Thiram along with *Rhizobium* and phosphate-solubilizing bacteria (PSB) to enhance germination and nutrient uptake.

For weed management, Pendimethalin is applied pre-emergence and Imazethapyr post-emergence, along with hand weeding at 20 and 40 days after sowing (DAS). To manage sucking pests, seed treatment with Imidacloprid is effective, while collar rot can be controlled by treating seeds with Fluxapyroxad or a combination of Carboxin and Thiram. In case of defoliator infestation, spraying Indoxacarb, Lambda-cyhalothrin, or a Novaluron mixture is recommended. For rust and leaf spot diseases, fungicides like Hexaconazole or Pyraclostrobin should be applied. Additionally, 75% of the recommended dose of fertilizers (RDF) should be supplemented with 25 kg/ha of Mycorrhiza and *Rhizobium* seed treatment to enhance soil health and crop productivity.

Cotton (Bt & Desi)

Cotton cultivation thrives best in well-drained loamy to sandy loam soils. Prior to sowing, deep ploughing is essential along with the incorporation of 10-15 t/ha of farmyard manure (FYM) to enhance soil fertility. Sowing is typically carried out in June-July under rainfed conditions and during April-May for irrigated crops. Recommended spacing for normal planting is 90-120 × 30-45 cm, while under high-density planting (HDP) systems, 60 × 30 cm spacing is advised. Suitable varieties include Bt hybrids like G. Cot. Hy. 24 and 26, and desi types such as GADC-3 and GADC-4. The seed rate varies by type, with 1.5-2 kg/ha for Bt varieties and 10-12 kg/ha for desi types.

Stop irrigation 10-12 days before harvesting; harvest when 80-85% grains in the panicle are matured. Dry harvested grains properly to maintain quality and reduce post-harvest losses.

Groundnut

Use improved groundnut varieties like GJG-32 and GJG-22 with a seed rate of 100-120 kg/ha. Treat seeds with *Rhizobium* culture and appropriate insecticides to prevent white grub infestation. After the first rainfall, spray surrounding trees to control grub sources and minimize field infestation.

Sesame

For effective cultivation, land preparation should focus on creating well-drained, loamy soil through 1-2 ploughings followed by levelling. The ideal sowing period is from the last week of June to mid-July, immediately after the onset of the monsoon. Recommended varieties include Gujarat Til 1, Gujarat Til 2, and Gujarat Til 3. The seed rate should be 5-6 kg per hectare, with a recommended spacing of 30 cm × 10 cm (row × plant) to ensure optimal growth. Seeds should be treated with either *Trichoderma viride* at 4 g/kg seed or Carbendazim at 2 g/kg seed to prevent seed-borne diseases.

For fertilizer application, a basal dose of 40 kg N and 20 kg P₂O₅ per hectare should be applied at the time of sowing. Weed management can be achieved through interculture or hand weeding, which should be done twice-once at 15-20 days after sowing (DAS) and again at 30-35 DAS. Pest management for leaf roller or capsule borer involves spraying Neem oil at 2% or Spinosad 45 SC at 0.3 ml/L. To manage diseases like Alternaria leaf spot or Phyllody, infected plants should be removed and, if necessary, Mancozeb 0.2% should be applied. Water management is primarily rainfed, but light irrigation should be provided if there is a dry spell lasting more than 10 days, particularly during the flowering or pod-filling stages.

Cotton

For *Bt* cotton cultivation, select recommended hybrids suited to the region, such as GCHH 245, GCHH 177, RCH 2 BG-II, and JKCH 1947. The ideal sowing period is from mid-June to early July, coinciding with the onset of the monsoon. Land preparation should involve deep ploughing followed by harrowing to achieve good tilth. Before sowing, treat the seeds with Carbendazim or Thiram at 2 g/kg to prevent seed-borne diseases.



Pest Surveillance in Cotton

The recommended spacing for *Bt* cotton hybrids is 120 cm × 60 cm, ensuring a proper plant population of approximately 13,000-15,000 plants per hectare. Fertilizer application should include 240 kg of nitrogen, 60 kg of phosphorus, and 60 kg of potassium per hectare. Apply 25% of the nitrogen, along with the full dose of phosphorus and potassium, as the basal dose, and split the remaining nitrogen into 2-3 applications during the growing

season. Weed management should start with a pre-emergence spray of Pendimethalin at 1 kg a.i./ha immediately after sowing, followed by manual weeding at 20 and 40 days after sowing (DAS).

Intercropping with green gram, black gram, or cowpea during the early growth stages can provide additional income and improve soil health. For plant protection, monitor for sucking pests like aphids, jassids, and whiteflies, and spray Thiamethoxam at 0.3 g/l or Imidacloprid at 0.3 ml/l if necessary. Bollworm control can be managed using pheromone traps, and sprays of Spinosad, Emamectin benzoate, or Flubendiamide should be applied when pests reach threshold levels. Biocontrol agents like *Trichogramma* can be used where feasible. To prevent fungal diseases such as Alternaria and anthracnose, spray a combination of Carbendazim and Mancozeb. Ensure clean cultivation practices and remove infected plants to prevent the spread of disease. Avoid waterlogging during the monsoon, and provide protective irrigation during dry spells, especially during boll formation. Growth regulation can be enhanced by spraying 2% DAP or 1% KNO₃ at flowering to improve boll retention.

Sugarcane

For sugarcane cultivation, it is important to select region-specific, high-yielding, and disease-resistant varieties such as Co 86032, CoN 05071, CoN 09072, and Co 99004. The ideal planting time is from mid-June to July, especially under rainfed conditions. Land preparation involves deep ploughing followed by harrowing, levelling the fields, and ensuring good drainage to prevent water stagnation. Use 3-budded setts from healthy, disease-free canes, and treat them with fungicides like Carbendazim 0.1% and insecticides such as Chlorpyrifos 0.1% before planting. The recommended row-to-row spacing is 120 cm, with a sett-to-sett spacing of 30 cm to ensure proper aeration and sunlight. Use 300 kg of nitrogen, 75 kg of phosphorus, and 75 kg of potassium (K₂O) per hectare. Apply the full dose of phosphorus and potassium, along with one-third of the nitrogen, at planting. The remaining nitrogen should be applied in two equal splits, one at 45 days and the other at 90 days after planting.

Weed management can be achieved using pre-emergence herbicides such as Atrazine at 1 kg/ha or Metribuzin at 1.25 kg/ha. During dry spells, provide light irrigation, particularly at the formative and grand growth stages, to maintain adequate moisture for proper tillering. Earthing up should be carried out at 45-60 days after planting (DAP) for partial earthing and at 90-120 DAP for full earthing to promote root growth and prevent lodging. Release *Trichogramma chilonis* at 50,000/ha/week to control early shoot borers, and apply Chlorpyrifos at 2.5 L/ha with irrigation to manage white grubs. In case of top shoot borers, spray Chlorantraniliprole or Spinosad when pests reach threshold levels. To prevent red rot and smut, use disease-free setts and rotate with non-host crops such as rice or legumes, and promptly remove and destroy infected clumps. Intercropping with crops like green gram, black gram, cowpea, or groundnut during the early growth stages can enhance income and improve soil fertility. Additionally, retaining sugarcane trash in the inter-row spaces helps conserve moisture and suppress weeds.

For plant protection, control Heliothis with marigold planting, manage Sterility Mosaic with Propargite spray, and control Powdery Mildew with Sulfur or Carbendazim spray.

Tomato

For tomato cultivation, recommended varieties include Gujarat Tomato-2, Gujarat Tomato-5, and Anand Tomato-8. The seed rate is 200-250 g/ha, and the ideal spacing is 90x75 cm or 70x60 cm. For nursery management, apply Metalaxil MZ to prevent root rot and use micro-nutrient sprays. For plant protection, control Leaf Miner with Neem powder/oil spray, and use Spinosad for severe infestations. Manage Early Blight with Mancozeb or Chlorothalonil spray.

Chilli

For chili cultivation, recommended varieties include GVC-101, Anand Jwala, and hybrid varieties. The seed rate is 750 g/ha, with spacing of 60x60 cm. Apply a basal fertilizer of 50:50:50 (NPK) kg/ha. For plant protection, control Thrips with Neem powder/oil spray and use Spinosad for severe infestations. Manage Leaf Curl virus with Pyriproxyfen or Fenprothetin spray.

Brinjal

For brinjal cultivation, use recommended varieties: Gujarat Anand Oblong Brinjal 2 and hybrid varieties. The seed rate is 250-300 g/ha, with spacing of 90x60 cm. For nursery management, use *Trichoderma viridi* and water as needed. For plant protection, control Whitefly with Neem powder/oil spray, and manage Little Leaf by removing affected plants and applying Pyriproxyfen spray.

To improve milk yield in lactating animals, formulate a proper feed rich in essential nutrients. Consider using *Azolla* as a green fodder replacement to enhance protein intake. During rainy seasons, keep livestock indoors and ensure proper fodder storage. Vaccinate animals against FMD (Foot and Mouth Disease) and Brucellosis, along with other necessary vaccines. Maintain clean and hygienic animal sheds to prevent mastitis. Deworm animals with Fenbendazole, and control ectoparasites using Deltamethrin or Amitraz.

NORTH GUJARAT-KUTCHH

This region of the state includes the regions of Banaskantha, Patan, Mehsana, Sabarkantha, Aravalli, and Gandhinagar. The local agricultural university is Sardarkrushinagar Dantiwara Agricultural University (S.K.Nagar, Banaskantha). The region receives rainfall between 700-1000 mm annually and is characterized by loamy sand soil.

Groundnut

For groundnut cultivation, the latest improved variety GG 37 (Bunchy type) is recommended. The ideal sowing period is from June 15th to July 10th, with a seed rate of 100-120 kg/ha and a row spacing of 45 cm. Employ the FIR approach (Fungicide, Insecticide, Rhizobium) for seed treatment, using Mancozeb @ 2.5 g, Quinalphos 25 ml, and *Rhizobium* 10 ml/kg of seed to manage white grub and collar rot. Apply a fertilizer dose of 12.5:25:0 NPK kg/ha, along with 8-10 t/ha of FYM. Supplement with secondary nutrient sulfur at 20 kg/ha. For collar rot management, apply

Plant protections: For the management of pod borer application of chlorantraniliprole 18.5 SC @ 150 ml/ha followed by flubendiamide 480 SC @ 125 ml/ha is advised to minimize the larval population or Spinosad (150 mL/ha) at flowering and pod initiation. In incidence of blister beetles deep ploughing, field sanitation, light traps, and neem-based products (NSKE 5% or Azadirachtin 0.03% WSP @ 2500-5000 g/ha) can be used.

Disease management: Sterility mosaic virus can be controlled by using resistant varieties and removing infected plants early to limit virus spread. For vector control, spray Neem oil (5 ml/litre) or NSKE 5% every 15 days. Seed treatment with Imidacloprid 70 WS (5 g/kg seed) helps protect seedlings. Alternaria blight can be managed by spraying Mancozeb (2.5 g/litre) or Carbendazim (1 g/litre), planting on raised beds for better drainage, and using resistant cultivars. Phytophthora blight can be controlled by seed treatment with Metalaxyl (8g/kg), maintaining good drainage, minimizing stem injury, and using resistant genotypes.

Summer moong

If infestation of Jassid and white fly is observed, apply 1.0 litre malathion 50 EC or 625 dimethioate 30 EC per hectare in 625 litres of water. Harvest the crop at ripening of 70-80 percent pods to allow timely sowing of the kharif crop.

OILSEED CROPS

Groundnut

Improved Varieties: MH-4, HNG-69, GJG-19.

Time of sowing: Optimum time of sowing of groundnut is month of June. Under irrigated conditions, sowing should be completed by second fortnight of June, whereas under rainfed conditions sowing at the onset of monsoon should be done. Sowing of crop after 15 July should be avoided.

Seed rate: The seed rate depends on the size of kernels of the varieties. However, for medium size kernels varieties, seed rate of 80 kg per hectare is required.

Method of sowing: Sowing should be done preferably in rows at a spacing of 30 cm and maintaining plant to plant spacing of 15 cm.

Seed treatment: Treatment of seed with Thiram or Captan or Emisan 3 g/kg seed is essential for management of seed and soil borne diseases. For management of termites and white grub treat the seed with 15 ml Chloropyriphos 20 EC/kg seed.

Fertilizer use: Application of 15 kg N, 50 kg P₂O₅, 25 kg K₂O and 25 kg Zinc sulphate per hectare is required for obtaining good crop yield. It will be desirable to apply doses of all the nutrients before sowing. Use of gypsum as sulphur source has been found beneficial in groundnut crop.

COMMERCIAL CROP

Cotton

Time of sowing: Sowing of the crop is done from April to first fortnight of June. However, in case of American cotton optimum time of sowing is May. Delay in sowing



Management of paddy field

per hectare within 1-2 days of sowing. Bispyribac 10 SC should be applied at 250 ml/hectare 20-25 days after sowing. In case of chlorosis (iron deficiency), spray 1% ferrous sulphate solution (2.5 kg in 250 litres of water) two to three times at weekly intervals, and ensure adequate irrigation. For Terai paddy, seedlings should be prepared as per medium land rice guidelines. Varieties like Rajendra Mahsuri, Rajshree, and BPT-5204 should be selected and purchased in advance, preferably in May.

Maize

Farmers are advised to plough the land during May and apply 3-4 quintals of lime per hectare. Seeds should be sown in the first week of June or at the onset of the monsoon. To control termite populations and larvae, 100-150 quintals of well-decomposed farmyard manure (FYM) along with Methyl Parathion 2% dust at the rate of 25 kg per hectare should be incorporated into the soil during land preparation. Use 20 kg of seed per hectare of recommended maize varieties such as PMH-3, Orange Flint, HQPM-1, or HQPM-5. At the time of sowing, apply 86 kg of urea, 130 kg of DAP, and 68 kg of MOP per hectare. If the field remains dry, provide irrigation after 30 days. After proper weeding, apply an additional 65 kg of urea. Ensure the field has proper drainage to prevent water stagnation. Two weeks after germination, apply 10-15 granules of Fipronil 0.3G or Carbofuran 3G in each plant to manage early-stage pests. To control the incidence of Fall Armyworm in the maize crop, spray Emamectin Benzoate 5% SG at the rate of 1 gram per 3 litres of water.

PULSE CROPS

Farmers are advised to grow pulses (especially pea, chickpea, and lentil) during Kharif season. The recommended varieties for upland field include: Pigeon pea- Narendra Arhar-1, Narendra Arhar-2, Rajeev Lochan, IPA 203, Malviya Arhar-Mal-13; Black gram- WBU-109, Birsa Urd -1, PU-31; Green gram- Virat, Shikha, HUM-12, IPM 2-14, IPM 2-3, HUM-16; Horse gram- Birsa Kulthi-1, Payur-2, Madhu, VLG-19 etc. Oil seed crops like Niger- Utkal Niger-150, DNS-4, BNS-3, Pooja, JNS-29, JNC- 6, BNS-1; Sesame-RT-351, Krishna, Sekhar, RT-346, GT-4, GT-6, Punjab-1; Groundnut- K1812, TLG-45, ICGV-00350, K-6, TG-37A, TG-38 Dharni, Grinar-3, TG-51, TCGS-1043, AK12-14 etc.

Pulses (especially pea, chickpea, and lentil) suffer White powdery fungal growth on leaves, leading to leaf yellowing and reduced yield. Carbendazim 50% WP @ 1.0 g per litre of water (apply at first sign of disease) or Tebuconazole 25% EC @ 1.0 ml per litre of water (apply preventively or when symptoms first appear) is advised. Blight (*Alternaria*, *Cercospora* spp.) affect chick pea, use of Mancozeb 75% WP @ 2.5-3.0 g per litre of water (apply when symptoms first appear) and copper Oxychloride 50% WP @ 2.5-3.0 g per litre of water (apply at early stages of disease) is advised. Aphids (*Aphis craccivora*) affect pulses especially chick pea, use Imidacloprid 17.8% SL @ 0.5 ml per litre of water or Thiamethoxam 25% WG @ 0.3 g per litre of water. For Podborer (*Helicoverpa armigera*) spray Spinosad 45 SC @ 0.3 ml/litre or Emamectin Benzoate 5 SG @ 0.5 g/litre at flowering and pod formation stages.

OILSEED CROPS

Farmers should select light sandy soil with good drainage for cultivation, and land preparation should begin in the first week of June. Recommended varieties for groundnut include K-1812, K-6, Dharni, Birsa Mungfali-3, Birsa Mungfali-4, and Birsa Bold. For niger, recommended varieties are Birsa Niger-1 and Birsa Niger-3. These seeds should be purchased in May and sowing should be completed before July 10. For sesame, suitable varieties include RT-351, Krishna, Kanke Safed, and Shekhar. For soybean, recommended varieties are Birsa Safed Soybean-2 and JS-2098, and for sunflower, DCS-107 is suggested.

Before sowing, seeds should be treated with carbendazim at 2 g/kg of seed and chlorpyrifos at 6 ml/kg of seed. At the time of sowing, fertilizer should be applied at the rate of 25:50:20:20 kg per hectare of N:P₂O₅:K₂O:S (Nitrogen: Phosphorus: Potash: Sulphur). For termite control (*Odontotermes* spp.), pre-sowing treatment should be done with Chlorpyrifos 20% EC at 2.0-2.5 ml per litre of water as a soil drench. For post-sowing treatment, apply Fipronil 0.3% GR (granular formulation) at 10-12 kg per hectare to control active termite populations. In case of rust (*Puccinia arachidis*), Propiconazole 25% EC should be used at the rate of 1 ml/ L of water as a foliar spray during early flowering or upon symptom appearance. To prevent damage caused by the Bihar hairy caterpillar, affected leaves should be plucked and buried in pits at the initial stage. If infestation worsens, Indoxacarb 15.8 SC should be sprayed at the rate of 10 ml per litre of water. For management of tikka disease or brown spots on leaves, Dithane M-45 should be sprayed at 1.25 kg /ha.

FRUIT CROPS

Mango

Mango varieties such as Amrapali, Dussehri, and Neelanchal Kesari should be irrigated at weekly intervals to ensure proper fruit growth and development. For quality fruit production, apply 500 grams of urea and 400 grams of muriate of potash per plant. During the fruit development stage, spray micronutrients on the leaves. After harvesting, provide adequate nutrition to the plants to restore their vitality. With the onset of the monsoon, sow green manure crops such as 20 kg of hemp or 10 kg of dhaincha per acre. Incorporate manure and compost into the soil 45 days after sowing. Pruning of mango trees should be carried out in the months of December and January.



In-situ moisture conservation measures covering both soil and land management practices to be practiced. They include field bunding, compartmental bunding, ridges and furrows, conservation furrows, broad bed and furrow systems, dead furrows at 30 DAS especially in red soils for groundnut, groundnut+redgram, finger millet + pigeonpea, pigeonpea + field bean and pigeonpea + cowpea intercropping. Frequent interculture required both in kharif and rabi for weed control and moisture conservation. Avoid dry sowing of crops such as cotton, groundnut especially in shallow soils. Sowing needs to be done only when sufficient moisture is available in soil root zone. Unless 50 - 60 mm rainfall is received over 2-3 days, sowing should not be carried out. Facility of crop insurance including weather-based insurance to be availed. Follow the weather advisories available for the region. Do not burn crop residue. Take up rotavator operation in harvested fields of Cotton, Wheat and Safflower for incorporation of stubbles and crop residue in the soil. Border crops (Jowar/Bajra/Maize) should be practiced in vegetable growing fields which will prevent the vector entry in the main field. Attend ploughing operation immediately after the harvest of rabi crops which reduces the pest & weed incidence in next season.

CEREAL CROPS

Paddy

Paddy is commonly cultivated using a seed rate of 62.5 to 75 kg per hectare, depending on the variety. Popular varieties include MO-4, Abhilasha, Intan, RNR 15048, Tunga, KHP-10, and BPT 5204. Sowing is generally done in June to August, with some varieties like RNR 15048 and BPT 5204 sown in the first fortnight of July and transplanted in August. The recommended fertilizer dose for most varieties is 75:75:90 kg of N:P:K per hectare, while BPT 5204 and RNR 15048 require a higher nitrogen input of 150 kg along with 75:75 kg of phosphorus and potassium. Seed treatment is a crucial practice and involves the use of Carbendazim 50 WP at 2 grams per kg of seed to prevent fungal infections. In some cases, seeds are also treated with

Groundnut

Recommended varieties are Dh 256 and Kadarilepaxi, sown during the rabi/summer season at a seed rate of 150 kg/ha. Fertilizer application includes 18 kg N, 46 kg P₂O₅, and 25 kg K₂O per hectare, along with 500 kg gypsum and 25 kg each of zinc and iron sulphate. Seeds are treated with 30 g of Captan 80 WP before sowing, and dibbling is done at a spacing of 30 x 10 cm. No intercrop is recommended. Foliar application of 20 g urea per litre of water is done at 30 and 60 days after sowing. Irrigation is crucial at flowering and pod development stages.

Inter-cultivation is carried out at 15, 30, and 40 days after sowing. Weed management includes pre-emergent application of 1.3 litres of Pendimethalin 30 EC. Major diseases include leaf spot, managed by spraying 1 g Carbendazim 50 WP or 1 ml Hexaconazole, and collar rot, managed by seed treatment with 3 g of Carbendazim 25% + Mancozeb 25%. Pest control includes spraying 2 ml of Profenofos 50 EC or 1 ml of Monocrotophos 36 SL for leaf miner, and 2 ml of Quinolophos 25 EC for Spodoptera caterpillar. Application of 2 quintals of neem cake before sowing is also advised.

PULSE CROPS

Redgram

Recommended varieties of redgram include TS-3R, GRG-811, and GRG-152. Use 10-12.5 kg of seed per hectare and sow during June-July. Apply fertilizers at 25:50:75 or 25:50:125 kg NPK per hectare in red soils and 25:50:0 in black soils. Treat seeds with *Rhizobium* 500 g, PSB 500-1250 g, PGPR 500 g, Trichoderma 10 g/kg, and soak in 2% calcium chloride solution followed by shade drying for 7 hours. Use a spacing of 90 x 30 cm under normal planting, 120 x 20 cm for wider rows, and 45 x 10 cm for compact planting. Redgram can be intercropped with bajra, little millet, greengram, soybean, or blackgram. Apply pulse booster or pulse magic during flowering and pod-filling stages.

For weed management, use Pendimethalin at 3.25 L/ha as pre-emergence and Imazethapyr at 1 L/ha at 3 weeks after sowing. For disease management, use *Trichoderma*-treated seeds for wilt, Oxydemeton-methyl at 1.5 ml/L for sterility mosaic, and Metalaxyl + Mancozeb at 4 g/kg seed for *phytophthora* blight. To control pod borers, use Emamectin benzoate at 0.3 g/L or Coragen at 0.3 ml/L. For *Helicoverpa*, apply Profenophos at 2 ml/L, NSKE at 3 ml/L, or Chlorantraniliprole at 0.15 ml/L. Practice nipping at 45-50, 55-60, and 75-80 days after sowing to promote branching.

Greengram

Recommended varieties for greengram include DGGV-2, BGS-9, Pusa Baisakhi, and DGGV-1. The seed rate varies from 7.5 to 15 kg/ha, and sowing is done during June-July. Apply fertilizers at 12.5:25:20 or 25:50:0 kg N:P:K/ha as a basal dose at sowing. Treat seeds with *Rhizobium* 500 g and PSB 1250 g. Maintain spacing of 30 x 10 cm or 37.5 x 10 cm. It can be intercropped with redgram or sugarcane. During flowering and pod formation, spray pulse booster or pulse magic at 8-10 g/l.

For weed control, apply Pendimethalin at 3.25 l/ha immediately after sowing.

as banana, cocoa, nutmeg, black pepper, or ginger. Fertilizer recommendation is 100:40:140 g NPK per plant annually, applied in two equal splits - first in September-October along with 12 kg organic manure and second in February (for irrigated) or June (for rainfed). Additionally, zinc sulfate and borax are applied at 5 g per plant during October-December. Irrigation is usually done through the basin method at 175-200 litres per palm per irrigation, adjusted as per season: every 7-8 days in November-December, every 6 days in January-February, and every 4-5 days from March to May. Drip irrigation is also practiced. Common diseases include bud rot, leaf spot, collar rot, crown rot, foot rot, and yellow leaf disease. Bud rot can be managed with sprays of 1% Bordeaux mixture, 0.3% Copper Oxychloride (COC), or 0.2% Metalaxyl + Mancozeb 72WP, applied before the onset of monsoon and repeated after 30-45 days. Pests such as spindle bug, Pentatomid bug, red palm weevil, mites, root grubs, inflorescence caterpillar, and ambrosia beetle affect the crop. Red mite infestations are managed by spraying wettable sulphur (80%) at 0.4%.

LIVESTOCK

To ensure the health and productivity of livestock during the kharif season, it is essential to implement appropriate management practices. Cattle, buffalo, sheep, and goats should be vaccinated against Hemorrhagic Septicemia, while sheep and goats also require vaccination against Enterotoxaemia. Deworming of all animals using Piperazine, Albendazole, or Fenbendazole is recommended to control internal parasites during the rainy season. Along with green fodder, daily feeding of dry fodder to cattle, buffalo, and sheep is necessary for better digestion and nutrient absorption. Grazing of maize and jowar seedlings should be strictly avoided, as these can cause cyanide poisoning, potentially leading to animal death.

Pesticides, insecticides, and fertilizers must be stored safely away from animals to prevent accidental toxicity. Farmers can prepare animal feed using locally available ingredients, such as cereals (maize, jowar, wheat) 35-40 kg, bran or polish of grains 30-35 kg, oil cakes (groundnut, safflower, cottonseed) 25-30 kg, mineral mixture 2 kg, and salt 0.5-1 kg to make 100 kg of cattle feed. To prevent the spread of Lumpy Skin Disease (LSD), which is a vector-borne illness, livestock should be protected from mosquitoes, house flies, and ticks by using mosquito nets, repellents, and washing animals with ectoparasitocidal drugs. Additionally, burning dry neem leaves in the morning and evening can help reduce mosquito bites. Farmers are also encouraged to cultivate improved fodder and crops during the Kharif season, utilizing the monsoon rains. Furthermore, all backyard poultry birds should be compulsorily vaccinated against Ranikhet Disease.

Fisheries

Maintain 6 feet water in the fish pond for proper growth and development of fishes. Remove unwanted plants and animals prior to release of fishes in the pond. As a supplementary food, floating pallets or groundnut cake (25%), paddy husk (40 %), sweet potato powder (10%), fish meal (24 %), vitamin and mineral mixture (1%) should be mixed and feed scientifically. Continue to feed fishes 3-5 % of body weight on daily basis. The suitable breeds for fresh water aquaculture are catla, rohu, mrigal, amur common carps, jayanti rohu, punggiasius, GIFT tilapia, grass carp, silver carp, murrel. Prior to release of fingerlings (15 days) mix cow dung (100 kg/100 m²) in the pond for plankton production.

Weed control is achieved by mulching with banana leaves or straw, which also helps in moisture conservation. Intercropping with legumes like cowpea or groundnut is sometimes practiced. Drainage channels are constructed to avoid water stagnation during heavy rains.

Pest and disease management is critical during the rainy season. Pseudo-stem and rhizome weevils are managed using chlorpyrifos or neem-based insecticides, while Sigatoka leaf spot is controlled by removing affected leaves and spraying fungicides such as carbendazim or mancozeb. Proper drainage and field sanitation help reduce the incidence of Panama wilt. To prevent lodging due to strong winds and heavy rains, tall varieties are supported using bamboo props or coir ropes. De-suckering is done regularly to maintain one mother plant and one follower per pit, while deleafing involves the removal of dried or diseased leaves to reduce humidity and disease pressure.

SPICE CROPS

Black Pepper

Field sanitization: Phyto-sanitary measures to be done which includes the removal and destruction of dried and diseased vines, fallen leaves, and any plant debris that may harbour pathogens. Thoroughly removing the litter from the base of pepper before the rains will prevent the diseases such as foot rot.

Shade regulation: Before the onset of monsoon, the unnecessary branches of the supporting trees should be cut and replaced to provide shade. Prune the runner shoots or tie back to the standards before the onset of the monsoon. Pruning can be repeated in July-August in over shaded gardens. It will help the gardens to get good air circulation and sunlight and reduce the attack of pollu beetle.

Intercultural Operations: With the onset of monsoons inter cultural operations in the interspaces of the pepper vines helps to improve soil aeration and water infiltration. This not only supports root development but also facilitates better microbial activity in the soil, contributing to overall plant vigor.

Soil Acidity Management: It is a critical factor in pepper cultivation, especially in the lateritic soils of Kerala that tend to be acidic. Application of lime @ 500 grams per vine need to be done after receiving rainfall. It will help to reduce the incidence of fungal diseases too. Liming helps to neutralize soil acidity, enhances nutrient availability, and promotes better uptake of essential elements by the plant roots.

Organic manure preparation: Prepare *Trichoderma* enriched compost to apply around the vines at the rate of about 10 kg/vine. Dry neem cake (10kg) and cow dung (90kg) are to be powdered and mixed to get a coarse texture and then moistened by sprinkling of *Trichoderma spp.* @ 1-2 kg per 100 kg of neemcake - cow dung mixture. After thoroughly mixing, cover it with a perforated polythene sheet or ordinary newspaper and keep it in shade for 4-5 days for multiplication. Again mix well and keep for three more days for further multiplication. This preparation is ready for incorporation in the soil. Application of *Trichoderma* enriched organic manure prevent the diseases like quick wilt in black pepper.



The Kharif season, stretching from June to October, is a crucial agricultural period for the state of Maharashtra, significantly influencing the livelihoods of millions of farmers. Characterized by the onset of the southwest monsoon, this season supports the cultivation of key crops such as rice, cotton, soybeans, pulses, and millets. However, climate variability, erratic rainfall, and changing weather patterns pose continuous challenges to sustainable agricultural production. In the light of these challenges, the Kharif 2025 Agro-Advisory aims to provide timely, region-specific, and science-based recommendations to empower farmers with the knowledge needed to make informed decisions.

REGIONAL ADVISORY

Maharashtra is divided into several agroclimatic zones, each with unique characteristics influencing crop selection, sowing windows, and input management. The following region-wise advisories aim to support farmers with tailored recommendations for improved productivity and resilience.

VIDARBHA REGION

The Vidarbha region of Maharashtra is composed of Bhandara, Chandrapur, Gondia, Gadchiroli, Nagpur, Amravati, Akola, Yavatmal, Buldhana, Wardha, Washim and possesses state agricultural university viz., Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola.

Soil Testing & Preparation (All Crops)

Soil testing is crucial for effective fertilizer management, optimizing input use, and maintaining soil fertility. For proper soil sampling, the recommended depths are 15-20 cm for crops like sorghum, rice, groundnut, and wheat; 30 cm for cotton, sugarcane, and banana; and 30 cm for fruit trees, sampled from a distance of 1-1.5 feet from the trunk. Additionally, applying decomposers to crop residues like straw and leaves enhances organic matter content, improves soil structure, and boosts water retention, contributing to sustainable agricultural practices.

Sugarcane

Sugarcane is planted during the Kharif season (June-July) with a duration of 12-15 months. It thrives best in well-drained deep loam soils with good water-holding capacity, avoiding saline or alkaline conditions. The climate should be warm and humid. Land preparation involves deep ploughing (1-2 times) followed by two cross harrowings, stubble and weed removal, and proper field leveling with drainage. Recommended varieties include CO-86032, CO-8014, CO-94012, VSI-434, and CO-M-0265, especially early-maturing ones. Healthy 9-10 month-old canes are selected for setts, taking the top two-thirds portion and avoiding flowering canes. The seed rate is 45,000-50,000 three-budded setts per hectare, approximately 6-8 tonnes. Before planting, setts are treated with 0.25% Bavistin or Carbendazim mixed with a 2.5% urea solution for 10-15 minutes.

Spacing varies: 90 cm for rainfed, 120 cm for irrigated or paired row trench planting, and 60 cm for High-Density Planting Systems (HDPS). Planting is done in furrows or trenches, with buds facing upward, covered lightly with soil. The fertilizer dose is 250:115:115 NPK kg/ha for irrigated crops and 200:90:90 NPK kg/ha for rainfed crops. At planting, 1/3 nitrogen (N) along with full phosphorus (P) and potassium (K) is applied, while the remaining N is split and applied at 45 and 90 days after planting (DAP). Fertilizers are placed in bands 8-10 cm deep and 10-15 cm away from the rows. Additionally, 25-30 t/ha of FYM or compost is applied before the final harrowing. Biofertilizers such as Azotobacter and PSB (2 kg/ha each) mixed with 100 kg of compost or FYM are added during planting. Intercropping with green gram, black gram, soybean, or groundnut is recommended during early stages. Light earthing up is done at 45-60 DAP, followed by full earthing up at 90-120 DAP. Trash mulching with cane trash after 60 DAP helps conserve moisture and control weeds.

For pest control, early shoot borer is managed by removing affected shoots, releasing *Trichogramma* (@ 50,000/ha/week), or applying Chlorpyrifos 20 EC (5 ml/L). Internode borer is controlled with Cartap hydrochloride 4G (10 kg/ha) or *Trichogramma* release. White grub management includes applying Phorate 10G (25 kg/ha) in moist soil. Red rot is prevented by planting resistant varieties, removing infected clumps, and treating setts with Carbendazim 0.1%. Smut is controlled by using disease-free setts, hot water treatment (50°C for 30 minutes), or fungicide dips. Wilt is managed through resistant varieties and crop rotation. For growth enhancement, 2% urea or 2% DAP is applied during tillering and grand growth stages if the crop appears pale or yellow. These practices ensure healthy growth and optimum yields.

PULSES

Soybean

Sowing of soybean is recommended between 20th June and 5th July, with a seed rate of 65 kg/ha and a spacing of 45 cm. Preferred varieties include MAUS-162, MAUS-612, PKV Amba, and PKV Yellow Gold. Seeds should be treated with a fungicide-insecticide combination like Azoxystrobin (10 ml/kg) and inoculated with Rhizobium and PSM (5 g/kg). For weed control, apply Pendimethalin 30% EC (3.25 L/ha) or Flumioxazin 50SC (250 ml/ha) as a pre-emergence herbicide, followed by

to prevent seed-borne diseases. In regions with heavy rainfall, wet bed nurseries are recommended, especially on low-lying lateritic or clay loam soils. Where drainage is poor, raised beds should be prepared. Select well-drained and fertile fields near water sources, avoiding waterlogged or saline areas. Apply 2 kg of urea and 1 kg of SSP per 100 m² of nursery, and incorporate well-decomposed FYM or compost (5-10 t/ha) during bed preparation.

Pre-germinated seeds should be uniformly broadcasted on nursery beds with light pressing to ensure proper soil contact. Maintain a shallow water level of 2-3 cm during the early growth stage, avoiding stagnation to prevent diseases. Weed pressure is generally low, but light raking or manual weeding should be done 10-12 days after sowing if required. To manage weeds effectively, spray Butachlor 50 EC @30 ml/litre water 2-3 days after sowing. Transplant 21-day-old seedlings with a spacing of 20×15 cm for normal varieties and 15×15 cm for short-duration varieties. At transplanting, apply 10 t/ha of Gliricidia leaves or 8 t + 50 kg of nitrogen for better puddling and root establishment. Give the second nitrogen dose (1 kg urea/guntha) 15 days after transplanting to boost growth. Replant damaged or gap areas promptly with healthy seedlings to maintain uniformity

Fertilization and Pest/Disease Control: Apply 3 kg of compost per m² of nursery bed before sowing. For mechanized farming, mat nurseries on plastic sheets or trays are recommended, with fertilizers applied at 2.2 kg urea, 3 kg SSP, and 0.8 kg MOP per guntha. For effective pest control, monitor the nursery for stem borers and use Quinolphos (150 ml/guntha) or Carbofuran (165 g/guntha) as needed. Crab control can be managed using toxic bait prepared with Acephate or Carbaryl mixed with cooked rice. Monitor for diseases like blast, brown spot, and root rot, and apply fungicides and insecticides as per local recommendations.

MILLET CROPS

Finger Millet (Nagli) & Proso Millet (Vari)

For effective nursery preparation, sow the rice seeds in rows on 10 guntha nursery beds using a seed rate of 6 kg per hectare. Fifteen days after sowing, apply Quinolphos at 15 kg/ha or Carbofuran at 16.5 kg/ha to manage early pest infestations. To prevent blast disease, spray Zineb at 2 kg or Copper Oxychloride at 1.25 kg mixed in 500 liters of water per hectare. Transplant healthy 4-5-week-old seedlings at a spacing of 15×20 cm, and ensure proper nutrient application with a recommended dose of 40:40 N:P₂O₅ kg/ha to promote vigorous growth and better yield.

OILSEED CROPS

Groundnut

Incorporate 5-7.5 t/ha of well-decomposed FYM during the final tillage to enhance soil fertility. At the time of sowing, apply 25 kg of Nitrogen and 50 kg of Phosphorus per ha using boronated SSP for better root development. Use 100-110 kg of seed per hectare from recommended varieties like Konkan Bhuratna and TAG-24. For effective disease control, treat seeds with Thiram at 5 g/kg along with Rhizobium culture at 25 g/kg before sowing. Follow a sowing pattern of 30 cm paired rows with

60 kg N, 30 kg P_2O_5 , and 30 kg K_2O per hectare for optimal crop nutrition. Millets can be intercropped with soybean. Conduct three hoeings with a hand hoe during early growth stages for effective weed suppression. Spray Mancozeb (0.2%) at early crop stages if Blast, Brown Spot, or Rust symptoms appear. Rogue out diseased plants to manage spread and for armyworms, cutworms, and leaf-scraping beetles, apply Thiamethoxam @ 5 g/15 litres of water for effective protection.

PULSE CROPS

Blackgram

Conduct deep summer ploughing to expose and destroy the eggs, larvae, and pupae of major pests like pod borer, caterpillars, and weevils. Mix FYM/compost into the soil 10-15 days before sowing. Ensure the field is free from weeds for optimal crop growth. Use high-yielding varieties like PU 31 and LBG-20 (Teja). PU 31 is tolerant to Yellow Mosaic Virus (YMV) and well-suited for Kharif cultivation, while LBG-



Varietal demonstration in blackgram

20 performs well in all seasons. Use a seed rate of 10-15 kg/ha (or 2.5-3.75 kg/sangam). Sowing can be done by broadcasting or dibbling at a spacing of 15 cm × 30 cm. If required, apply lime in the furrows at sowing, preferably every alternate year, based on soil test recommendations to manage soil acidity. Treat seeds with Carbendazim @ 2g/kg of seed before sowing to protect against seed-borne diseases and promote healthy germination. Apply 50 kg potash/ha to the crop raised with closer spacing (20X10cm) to reduce stem fly and pod borer incidence. Treat the seeds with Carbendazim 50WP or Thiram @ 2g/kg of seeds or with Trichoderma viride @ 4g/kg of seeds before sowing. Install yellow sticky traps @ 10 traps/ha in the field to manage aphids. Spray Dimethoate 30EC @ 1ml/litre of water or Imidacloprid 17.8SL @ 0.5ml/litre of water for managing sucking pests.

OILSEED CROPS

Groundnut

Perform one ploughing with a soil-turning plough followed by two harrowings to achieve fine tilth up to 12-18 cm. Sow from 1st week of May to 1st week of June for optimal growth. Use ICGS 76 (semi-spreading, matures in 120-125 days) and JL 27 varieties. Treat groundnut seeds with Dithane M-45 or Thiram @ 3 g/kg to protect from seed and soil-borne diseases. For both soybean and groundnut, seed treatment is crucial under dry conditions. Keep the groundnut field weed-free for up to 50 days after sowing, as weed pressure is high due to monsoon rains. Timely inter-cultivation and weeding are essential for higher yields. Perform earthing up at 30-35 days after sowing in groundnut to promote pegging and pod development. Promote groundnut and soybean in foothill areas with limited irrigation, where rice cannot be grown, as climate-resilient and economically viable alternatives.

local short duration var. Idaw, tai, Buhsakei are recommended for upland/Jhum cultivation with rich alluvial soil

Pest & Disease Management: To effectively control Gundhi bugs, deploy 100 rotten crab funnel traps per hectare during the critical milking stage of the crop. For organic pest management, utilize neem-based biopesticides, release *Trichogramma* egg parasitoids, and install yellow sticky traps to monitor and reduce pest populations. Additionally, conserve natural predators such as spiders, dragonflies, and ladybugs by minimizing chemical pesticide use and maintaining ecological balance in the field. Adopt SRI method in small areas for enhanced yields, conduct regular field monitoring for pests and nutrient deficiencies and maintain proper water management and spacing as per cultivation method.

OILSEED CROPS

Soyabean

Varietal Selection: RCS1-1, RCS1-9, RCS1-10, JS335, MAUS 71 is suitable option for Kharif cultivation in Mizoram.

Irrigation: Despite it is a rainfed crop in kharif, Supplemental irrigation is required depending on rainfall patterns. Flowering and pod filling stages are the critical stages of irrigation.

Relay cropping: Soybean can be grown as a relay crop with maize, planting between the rows of the maize crop is recommended.

Integrated nutrient management: Soybean being a leguminous crop does not require high dose of nitrogen, however 20 kg/ha is sufficient for healthy crop stand. A dose of 60 kg P_2O_5 (SSP) and 30kg K_2O (MOP) are recommended. FYM @ 5t/ha should also supplement the chemical fertilizers.

Integrated disease management: Seed treatment with Thiram at the rate of 4.5g per kg of seed will control fungal diseases like Leaf Blight, Leaf Spot, Seedling Rot, Frog eye spot and Rust.

Integrated pest management: Spray application of Nuvacron 1.25ml/L or Dimethoate (0.04%) can be used for effective control of leaf folder, stem fly and semilooper like pests. Seed treatment with Imidachloprid @7g/kg of seed reduces the problem of Leaf folder

Groundnut

For optimal groundnut production, farmers should complete sowing between 15th May and first week of June, using high-yielding varieties like ICGS 76, ICGV 86590, Girnar 4, TS 67 Before sowing, treat seeds with Bavistin (5g/kg seed) for disease prevention and apply Rhizobium + PSB culture for better plant growth. Prepare fields by incorporating lime to address soil acidity and use Butachlor (10ml/litre water) as a pre-emergence weedicide. Apply balanced fertilization at 20:60:40 kg NPK/ha, split into basal and top-dressing applications. To protect crops, spray Dichlorvos (2ml/L) against Blister beetles during flowering and implement monthly preventive sprays of Dichlorvos (2ml/litre) + Bavistin (2g/L) from 30 days after sowing. Ensure light

Groundnut

Sowing of Groundnut should be completed from fortnight of May to June first week using high yielding variety like ICGS 76, Dharani, KDG-123 and Girnar 4. Before sowing, treat seeds with *Rhizobium* and *PSB* culture for better growth and yield, and control blister beetles on groundnut flowers by spraying Dichlorvos @ 2 ml/litre of water.



Rhizobium and PSB culture in groundnut

PULSE CROPS

To manage pest infestations, collect and destroy skeletonized leaves, egg clusters, and larvae of *Spodoptera litura* and hairy caterpillars; use light traps or bonfires to control adult moths; intercrop with pearl millet or sorghum; apply entomopathogenic fungi like *B. bassiana* or *M. anisopliae* @ 5 g/L; and spray neem oil @ 5 ml/L to control aphids.

SPICE CROPS

Ginger

The rhizome should be sown by 1st to 3rd week of May in the fields with a spacing of 25 cm row to row and 25 cm plant to plant. Earthing up should be done during July to protect Rhizome fly infestation.

Management of Rhizome Rot Disease: Maintain raised beds or ridges and ensure proper drainage to prevent water stagnation and water logging. Apply mulch (green leaves/straw) to retain soil moisture and suppress pathogens, incorporate neem cake (2-5 tons/ha), apply *Trichoderma harzianum* to soil at 10-15 day intervals during early crop stages, and spray *Pseudomonas fluorescens* @ 5 g/L every 15 days for effective disease management.

Turmeric

Sowing of turmeric var. Megha turmeric 1 or Lakadong should be completed immediately with seed rate of 15-20 q/ha and 30*30 cm spacing. Earthing up should be done during July to protect Rhizome fly infestation. Planting in ridges and furrows and provide proper drainage coli, lettuce etc.



Sprouted rhizome ready for transplanting



Turmeric rhizomes

5204 are suitable. For coastal saline soils, farmers are advised to grow salt tolerant varieties such as CR Dhan 405, CR Dhan 403, CR Dhan 412 and Lunishree. Suitable varieties for upland situation are-Khandagiri, Vandana, Nabanna, Kolab, CR Dhan 202 and Sahbhagi dhan. Use herbicides for control of weeds for which pre-emergence herbicides such as Pretilachlor with safener@1500 ml/ha, pyrazosulfuron@200 g/ha, Pretilachlor 6% + Bensulfuron methyl 0.6% GR @ 10 kg/ha at 0-3 days of sowing or transplanting and post-emergence application of Fenoxaprop-ethyl @ 1300ml/ha + Ethoxysulfuron @ 100 ml/ha, Bispyribac sodium @250 g/ha, Triafamone + Ethoxysulfuron@90 g/ha, cyhalofop butyl+penoxsulam@2250 ml/ha at 15-20 days after transplanting/sowing may be used. Algal bloom is a major problem in transplanted rice. To control it, spray Copper oxychloride@ 3 g/litre of water or apply 10 kg CuSO₄/ha in soil. Install pheromone traps with scirpolure (5 no./ha) to monitor incidence of yellow stemborer. Tricho cards (50000 eggs/ha releas for 4 times) are recommended to manage the pests like yellow stemborer and leaf folder.

PULSE CROPS

Use suitable varieties of green gram- Virat, Sikha, IPM 2-14; black gram- PU 9, Sashi and IPU 2-43; and arhar- LRG 52 and BRG 2. For management of yellow mosaic vein virus in green gram/black gram, seed treatment with Thiamethoxam 30 FS or Imidachloprid 600 FS @5 ml/kg seed, installation of yellow sticky traps 50 no./ha 30 DAS along with need-based spray of neem oil 1500 ppm @1500 ml/ha alternately with Flonicamide@120 g/ha may be undertaken. Treat the seeds with *rhizobium* and *PSB* biofertilizers @25g each/kg seeds along with ammonium molybdate@ 3 g/10 kg seed mixing with jaggery and water before planting. To manage weeds Pendimethalin 30% as pre-emergence herbicide @2500 ml/ha or Quizalofop ethyl 5% @1000 ml/ ha as post-emergence herbicide may be used.

OILSEED CROPS

Groundnut

Cultivate suitable groundnut varieties like Kadiri Lepakshi, Dharani and Smruti. Soil application of 20 kg N, 40 kg P₂O₅, 40 kg K₂O/ ha at sowing and Gypsum @ 250 kg/ha at the time of hoeing should be applied to improve productivity. Tobacco caterpillars can be controlled with the spray Indoxacarb + Nuvaluron @ 625 per ha or Chlorpyrifos 50% + Cypermethrin 5% EC @ 1000 ml/ha or Fipronil 5SC @ 1000 ml/ha. Harvest the crop when plants turn yellow and leaves start drying. If no rain is received, apply light irrigation 2-3 days before harvesting. Dry the pods properly after harvest so as to reduce the moisture content to 8-9 % before storing.

COMMERCIAL CROPS

Sugarcane

Cultivate high yielding sugarcane varieties like Sabita, Nilamadhava and Charchika. Carry out wrapping and propping as the crop attains 4-5 months age in medium and late duration varieties. Spray Lamba cyhalothrin @2 ml/L of water to minimize infestation of top shoot borer. Avoid ratooning in red rot affected areas and sett treatment to be done with Azoxystrobin 18.2% + Difconazole 11.4% @ 0.1% per kg sett.

Rajasthan is predominantly a kharif cropping state due to vast area under rainfed cropping and crops. The diverse crops are grown during Kharif season. During May and June preparations for sowing of different crops like maize, bajra, green gram, cowpea, guar, moth bean are completed by the farmers. The sowing and nursery raising of cotton, groundnut and rice are done under varying situations and climatic conditions. To harness the potential of the crops, an appropriate technology along with management practices are inevitable. Therefore, a proper agro-advisory is need of the day to be issued for the use of the farmers for different agricultural operations. It will help in increasing the productivity and profit from agriculture and empower the farmers while helping in sustainable management of natural resources and applied inputs. The agro-advisories related to different activities and operations in agriculture conducted by the farmers during May-June are discussed in following para:

Soil Analysis

After the harvesting of rabi season crops in March and April and before sowing of kharif crops is the best time for taking soil samples from the farmers' fields for their analysis for physical, chemical and biological properties. The farmers are advised to take soil samples from 5 places of their fields, mix it properly and take them to nearby KVK or district soil testing labs for analysis. The sampling should be done from 0-30 cm soil depth. Attempt should be made get the soil sample analysed before sowing of kharif crops. The nutrient status as evidenced from soil sample analysis will help to decide right and balanced application of fertilizers and avoid extra application. It will also help in enhancing the nutrient use efficiency and saving of cost of cultivation while maximising the profit to farmers.

Summer Deep Ploughing and Application of Gypsum

May and June are the best time for summer deep ploughing and application of Gypsum in the field. The summer deep ploughing is process of solarization of the soil

profile which helps management and decomposition of crop residues, weeds, and unwanted plants in the soil. The deep ploughing in summer also kills the harmful insects, their eggs, larvae, and pupae as well as reduces the pathogens responsible for soil borne disease. Gypsum application is very important for Rajasthan soil which have by and large pH more than 8 and 8.5. The application of gypsum is decided based on the soil testing report. The farmers need to adequate quantity of gypsum which is decided on the basis of soil pH. Gypsum improves the soil reaction and brings to pH toward neutral range. It also helps in increasing the availability of plant nutrients such as sulphur and calcium. Gypsum is the cheapest source of sulphur and calcium. As Rajasthan is major R&M and Groundnut growing state, the increased availability of sulphur due to gypsum application will also enhance the yields of R&M in following season. The improvement in pH also helps in increasing the availability of all plant nutrients leading to higher production and quality improvement in kharif crops especially in oilseed and pulses.

Green manuring of Dhaincha

In salt affected soils alike of Rajasthan, green manuring with bulky organic matter crops like Dhaincha is the best option to improve the soil physical and chemical properties and soil fertility. This will also add nutrients in soil and help minimising over dependence on chemical fertilizers. Wherever irrigation facilities are available and farmers can afford to it, sowing of dhaincha for green manuring should be done in April-May. In rainfed areas, it should be done at the onset of monsoon. The seed rate of 60 kg/ha should be used for green manuring. The green crop of about 35-40 days stage (before flowering) should be ploughed down and incorporated in the soil with the help of rotavator or mouldboard plough. The optimum moisture in soil conditions should be maintained at the time of dhaincha incorporation in soil.

Arrangement of Quality Seed

Seed is the basic input for crop production. The farmers should arrange the quality seed of improved varieties or hybrids before the onset of sowing season. While purchasing the seed from any seed agency the labels of certification (foundation/certified/truthfully label) should be verified and ensured. Farmers are advised to contact and purchase quality seed from their nearby KVKs, Agricultural Research Stations, SAUs, different ICAR Institutes and State Seed Corporations and Cooperative societies. The new seed purchased authorised seed producing and marketing agencies are treated with fungicides and insecticides. Attempt should be made to purchase the treated seeds only. In case the seed is not treated, the same should be treated in the sequence of fungicide-insecticide -rhizobium (FIR) System before sowing.

Water Management

Rajasthan is a water-stressed state and water is the most critical natural resource for crop production. The land levelling using laser leveller in canal areas helps in enhancing uniform application of water wherever surface irrigation. This is the appropriate time for Laser land levelling. Farmers in canal network areas of Hanumangarh, Sriganaganagar and Hadouti Region of Rajasthan are advised to go for land levelling through laser land leveller during the months of May and June.

or 2 grams of Carbendazim per kilogram of seed. To control dry root rot, treat seeds with 2.0 g of Carbendazim or Thiophanate Methyl (70 WP) per kg of seed. Mix 10 g of *Trichoderma* per kg of seed along with 8-10 kgs of *Trichoderma* mixed with 400 kgs of moisture-rich farmyard manure. Keep mixture in shade for approximately 72 hours, and then apply it to the soil at a rate of per hectare.

Cluster bean

Time of Sowing: Under rainfed conditions, sowing should coincide with the onset of the monsoon, ideally within the first two weeks of July. Delays beyond first fortnight of July may lead to a noticeable decline in yield. In irrigated areas, sowing can be extended until the end of July.

Seed Rate & Spacing: Seed rate of 15 kg/ha is recommended, prior to sowing treat seeds with *Trichoderma* @ 4 g/kg or fungicides such as mancozeb or carbendazim @ 2 g/kg of seed, followed by chlorpyrifos @ 2 ml/kg of seed to deter termite attacks.

Improved Varieties: RGC 1038, RGC 1017, RGC 936, RGC 1002, RGC 1003, RGC 1033, RGC 1055, RGC 1066, Karan Guar 1 (RGr 12-1), Karan Guar 14 (RGr 18-1), and Karan Guar 15 (RGr 20-15). These varieties are suitable for semi-arid conditions and have high gum content.

Fertilizer Management: Application of 2.5 t/ha of compost or FYM 15 days before sowing. At sowing, apply 10 kg N and 20 kg P₂O₅/ha as basal application.

Water Management: Lifesaving irrigation during flowering and seed formation should be given if rainfall is deficient. Spraying with 0.1% thiourea at 25 and 45 DAS can boost yield under moisture stress.

Weed Management: Two manual weeding at 25 and 45 DAS are usually sufficient. Apply Pendimethalin (2.5-3.3 L/ha) within 2 days of sowing as pre-emergence, and Imazethapyr (400 g/ha) at 20-25 DAS as post-emergence for effective weed control.

Plant Protection: For termites' control in standing crops, mix Chlorpyrifos @ 1.25 L/ha or Fipronil 0.3% GR @ 20-25 kg/ha with irrigation water. Application of Imidacloprid, Dimethoate, or Malathion @ 0.75-1.25 ml/L of water effectively controls sucking pests like jassids, aphids, and whitefly.

Disease Management: To manage bacterial blight effectively spray with Streptocycline @ 5 g or Plantomycin @ 50 g in 100L water per ha at 35-40 days after sowing is beneficial. For *Angular leaf spot* and *Anthracnose* management, spray of Zineb @ 2kg in 500L water per hectare at 15-day intervals, at least twice is recommended. Powdery mildew can be controlled by spraying wettable sulphur (e.g., Sulfex) @ 2-3kg/ha, dusting sulphur powder @ 20-25kg/ha, or spraying Dinocap @ 1.5 ml/litre of water.

OILSEED CROPS

Groundnut

Time of Sowing: Rainfed Groundnut: Sowing with the onset of monsoon in the last week of June or in the first week of July. Irrigated Groundnut: Sowing to be done around 20th June or 10-12 days before the onset of monsoon with a pre-sowing irrigation.

Seed Treatment: Treat seed with fungicide Thiram @3 gm or carbendazim @1 gm + Thiram @2 gm or Mencozeb @ 2 gm per kg of seed. For control of termite seed treatment with chlopyriphos 20EC @ 4 ml per kg seed. For seed treatment *Trichoderma* @10 g or *Carbendazim* 50 WP @2 g or Tebuconazole 2 DS/kg @ 1.5 g seed kernel may be used before sowing for soil & seed borne disease. Seed should be inoculated with proper strain of *Rhizobium* culture particularly in those places where groundnut is to be grown for the first time.

Improved Varieties: GJG-37, RG-559, GJG-32, GJG-19, TCGS-1043 (Dharani), HNG-123, RG-425, RG-510, JL-501, Mallika (ICHG-00440), HNG-69, GJG-31, GJG-9, RSV-87, RS-138, JL-524, RG-141 are cultivated mostly.

Bio-fortified Varieties: Girnar-4 & Girnar-5 (Rich in Oleic acid).

Seed rate & Spacing: Spreading Type Varieties-80-100 kg/ha (45x10 cm); Bunch Type varieties-100-110kg/ha (30x10 cm)

Manure & Fertilizer: Apply FYM or compost @ 10-15 t/ha about 15-20 days before sowing. N:P: K @ 20:60:30 kg/ha. Basal application of fertilizers i.e., placing the fertilizers about 4-5 centimeter in the side of the seed and 4-5 centimeter below the seed at the time of sowing. Apply gypsum @ 375 kg/ha for proper development of pods and kernels.

Water Management: Pod development stage of groundnut is the most critical for water stress. Farmers should avoid moisture stress at this stage. In case drought or no rainfall, one irrigation should be given at pod development stage.

Weed Management: Major weed in Groundnut are *Amaranthus viridis*, *Digeria arvensis*, *Euphorbia hirta*, *Cyperus rotundus*, *Dactyloctenium aegypticum*, *Trianthema portulacastrum*, *Phyllanthus niruri*. To control these weeds following weed management practices should be adopted.

Pre Emergence: Pendimethalin 30% EC @1-2 Litre/ha, Oxiflorophen @ 0.25- 0.50 g a.i. per hectare and Diclosulam 22-26 g a.i. per hectare in 500-600 Litre water. Fluchloralin (Basalin) @ 1 kg a.i. per hectare dissolved in 800-1000 litres of water can also be used as pre-planting spray.

Post Emergence: Quizalofop-ethyl @ 0.050 g a.i. per hectare, Imazathayper @ 0.050 g a.i. per hectare in 500-600 Litre water.

Plant Protection: White gurb: Seed should be treated with Fipronil 5 SC @6 ml per kg of seed kernel and Clothianidin 50 WDG @1 g a.i. per kg of seed before sowing. Imidacloprid 17.8 SL @300 ml per hectare at 22 DAS. Biological Control by *Beauveria bassiana* @ 0.5 g/m² mixed with water in furrows at 15 days after sowing (DAS). Stem rot, collar rot, dry root rot, late leaf spot and rust: Seed treatment with Tebuconazole 2DS @1.5 g/kg of seeds or with *Trichoderma sp.* @10 g/kg of seed followed by furrow/soil application of *Trichoderma viride* @ 4 kg enriched in 250 kg FYM/ha. Thrips: Spray Malathion 5% powder @ 1 kg per hectare or dissolve 800 ml of Malathion 50 EC or 750 ml of Methyl Demeton 25 EC in water and spray. Bud necrosis: Treat the land with Carbofuran 3G @15 to 20 kg per hectare before sowing and spray Monocrotophos @1 ml per litre of water after 21 DAS. Termite: Apply Chloropyrifos @ 4 litres per hectare along with irrigation water.

Micronutrient Spray (Tomato, Bhendi, Chilli): TNAU multi micronutrient @ 1% during vegetative and flowering stage. Bhendi: Spray WSF 19:19:19 @ 2% + liquid Multi MN @ 1% at 30 DAS, repeated 3 times every 10 days.

Finger Millet

Organic Package: FYM @ 3 t/ha, Azophos (2 kg/ha), *Trichoderma* (2.5 kg/ha), vermicompost (1.5 t/ha), Panchagavya (3%) at vegetative, flowering, and grain filling stages. Disease/pest control: NSKE 5% and *Bacillus subtilis* 0.5% as needed. Foliar spray: 1.5% foliar formulation at flower initiation. Horsegram: Foliar Spray: TNAU Horsegram Wonder @ 5 kg/ha during flowering with wetting agent.

PULSE CROPS

Greengram, Blackgram, Redgram

Foliar Spray: TNAU Liquid Pulse Wonder @ 5 kg/ha at peak flowering (Greengram/Blackgram) and 15 days after (Redgram). Land preparation: Raised beds (90 cm) with 30 cm furrows. Seed Treatment: Imidacloprid (5 ml/kg) + *Bacillus subtilis* (10 g/kg) + *Rhizobium* & *Phosphobacteria* (30 g/kg). Agronomy: Spacing 30 × 15 cm; drip irrigation; fertigation 25:50:25:40 NPKS kg/ha. for weed control: Pendimethalin + Imazethapyr (PE) + Quizalofop-ethyl + Imazethapyr (EPoE). TNAU Vigour Plus: Nano-formulated seed coating (20-25 ml/kg) that boosts germination, seedling vigor, and early establishment. Bio-Pulse: Microbial seed bio-priming formulation that enhances yield under pathogenic stress, effective in chickpea, pea and lentil.

Blackgram

Nodule-Associated Plant Probiotics for Blackgram: Blend of beneficial microbes improving nodulation, drought tolerance, and yield (14% over standard). Newer fungicides for foliar disease management: Foliar spray of tebuconazole + trifloxystrobin (1g/L) or tebuconazole 25 EC (1ml/L), at symptom appearance and 15 days later; effective against *powdery mildew* and *Cercospora leafspot*.

Greengram

Newer fungicides for foliar disease management: Foliar spray of tebuconazole + trifloxystrobin (1g/l) or tebuconazole 25 EC (1ml/l), at symptom appearance and 15 days later; effective against powdery mildew and *Cercospora leaf spot*. TNAU Vigour Plus (Seed Booster): Pre-sowing seed coating with biopolymer matrix containing IAA and GA³; enhances seed germination and seedling vigour (also helpful in multiple crops like Pulses, Groundnut, Cotton, Paddy, Millets, Vegetables).

OILSEED CROPS

Groundnut

TNAU Groundnut Rich: A foliar nutrition solution. Applied twice @ 5.0 kg/ha at 35 DAS (50% flowering) and 45 DAS (pod development). Requires 500 litres of water per spray. Enhances pod development and improves yield. Nut Boost: A seed inoculation formulation containing a consortium of phosphate, potash, and zinc solubilizing PGPRs. Applied as a seed treatment. Improves pod yield by 16.5% to 18.1% compared to untreated seeds.

Greengram and Blackgram

Sowing time is generally from June 15th fortnight- July 15th during Kharif, Sept 15th to 30th October during Rabi, January last week to March 15th during Summer. Greengram varieties MGG-295, MGG-385, MGG-351, TM-96-2, WGG-37 and WGG-42, Virat, IPM-2-14, Sikha. Blackgram YMV-resistant varieties MBG-1070, PU-31, GBG-1 and TBG-104 are the preferred varieties. Treat seeds with imidacloprid @ 5g or thiamethoxam @ 5g/kg. Intercropping can be done with cotton (2:1 or 1:1 ratio) or redgram (6:1)/(8:1). Greengram can be taken as a preceding crop to Rice and Chilli as green manure for incorporation in the soil.

OILSEED CROPS

Groundnut

Seed rate recommended is 237 kg/ha and practice seed treatment with imidacloprid 600 FS @ 2 ml/kg followed by tebuconazole 2 DS @ 1g or Trichoderma viride @ 10g/kg to control early and late leaf spot diseases, sucking pests, collar rot, PSND in the early stages. Application of Gypsum @ 500 kg/ha at pegging stage near podding zone for pod development. Weed management with Diclosulam 84% WDG @31 gm/ha as pre-emergence herbicide followed by post-emergence application of Imazethapyr @ 750 ml/ha at 20 to 25 days after sowing.

Soybean

Sow varieties ASB-22, JS-335, AISb-1, JS-93-05 during 15th June 1st week of July using seed drill/ gorru/ desi plough/ by tractor drawn seed cum fertilizer drill / broad bed furrow planter. Sowing of Soybean by Broad bed Furrow (BBF) planter or Raised bed method facilitates the crop to withstand heavy rains and moisture stress conditions and to obtain higher yields. Treat the seeds with carbendazim @ 1g or thiram or captan @ 3 g/kg followed by 1.5 ml of imidacloprid (48 %) and Rhizobium culture @ 200 g/10 kg seed. Apply 50 kg DAP + 26 kg MOP at the time of sowing and top dressing of urea @ 25 -30 kg/acre at 30 DAS. Pre-emergence herbicides like pendimethalin 30 % @ 1 - 1.5 L/ acre should be applied within 24-48 hours after sowing with the help of hand sprayer for better control of weeds. Post emergence application of quizalofop ethyl @ 400 ml/ acre for control of grasses and imazethapyr + imazomax 40 g /acre or imazethapyr 10 % @ 250 ml/acre to control the grasses and broad-leaved weeds. To control stem fly and stem gridler as a prophylactic application of carbofuran 3G granules @ 8 kg/acre is recommended at 10 DAS. To control of pod blight spraying of tebuconazole + sulphur @ 2.5 g/L is recommended. Dry the seed and maintain the below 12 % moisture during seed storage to avoid incidence of pest and disease.

Castor

Sow high yielding castor hybrids viz., ICH-5, ICH-66, PCH-111, DCH-519 & ICH-6, PCS-262 from 15th June to 15th July in rainy season under rainfed conditions. Intercrop with redgram in 1:1 or greengram in 1:3 ratio, open furrows between rows to conserve moisture in light soils.

LIVESTOCK

Animal Husbandry & Fisheries

Agriculture and allied sectors play a vital role in the lives and livelihoods of the majority of Telangana's population. The state is rich in livestock resources and

cultivated in the field for the first time or after a long gap, to enhance nitrogen fixation. For weed control, one or two hand weeding should be done up to 40 days of sowing depending upon the weed intensity. Weeds can be controlled by the use of Pendimethalin 0.75-1.00 kg a.i./ha in 400-600 L of water as pre-emergence application.

OILSEED CROPS

Groundnut

Cultivate improved groundnut varieties such as Chitra, Kaushal, Prakash, Ambar, TG-37A, Utkarsh, and Divya, based on the length of the growing season and end-use requirements. Groundnut matures in about 120-130 days and should ideally be sown in the first fortnight of July to ensure harvest by the end of October.

Maintain row spacing of 45-50 cm and plant-to-plant spacing of 15-20 cm. Use 70-75 kg seed kernels/ha for timely sowing, and 90-95 kg/ha for late sowing conditions. Treat seeds before sowing with Thiram @ 2 g and Carbendazim @ 1 g per kg of seed, or Trichoderma @ 4 g + Carboxin @ 1 g per kg. Inoculate seeds with groundnut-specific Rhizobium culture 5-6 hours before sowing. Dissolve 50 g jaggery in 0.5 liters of water, mix with 250 g Rhizobium culture, and coat over 10 kg of seed. Dry treated seeds in the shade for 2-3 hours and sowing of seeds should be done in early morning or evening times to protect Rhizobium from heat-induced mortality.

Apply fertilizers based on soil testing; in the absence of test results, apply 20 kg nitrogen, 30 kg phosphorus, 45 kg potash, 250 kg gypsum, and 4 kg borax per hectare. Perform two hoeing-cum-weedings, the first at three weeks after sowing and the



Groundnut (TG 37A) at Mainpuri



Sesame (GJT 5) at Amethi

second three weeks later, as this helps minimize early weed competition, promotes better aeration, and supports healthy crop growth. For additional control, especially under high weed pressure, apply pre-emergence herbicide imazethapyr @ 100 g/ha in 500-600 liters of water within 3 days of sowing.

Soybean

Cultivate improved soybean varieties such as PK-472, JS-715, PS-564, PK-262, JS-2, JS-935, JS-7244, JS 20-29, Pusa-20, Pusa-16, and PK-416 based on growing season and production goals. The crop matures in about 120-125 days; sow between mid-June and the first fortnight of July. Maintain a row spacing of 45 cm and plant-to-plant spacing of 3-5 cm; sow seeds 3-4 cm deep. Use 70-75 kg seed/ha under normal conditions and 75-80 kg/ha under late sowing. Ensure a minimum germination of 75-80%.

Field Crops

India has one of the largest agricultural sectors worldwide, which is the primary source of livelihood for ~55% of India's population. India has the world's largest cultivated land under wheat, rice, and cotton. India is the largest producer of milk, pulses, and spices in the world. It is the second-largest producer of fruits, vegetables, tea, farmed fish, cotton, sugarcane, wheat, rice, cotton, and sugar. The agriculture sector in India holds the record for second-largest agricultural land in the world, generating employment for about half of the country's population. Thus, the farming community comprise an integral part of this sector to provide food security with a means of sustenance.

During the last 9 years (2014-2023), a total of 2279 varieties of field crops have been released in India under the aegis of ICAR, which include 1092 of cereals, 343 of oilseeds, 343 of pulses, 137 of forages crops, 279 of fibrecrops, 68 of sugarcane and 17 of potential (underutilized) crops. These varieties include 1888 climate resilient varieties (83%) tolerant to one or more biotic and/or abiotic stresses. Out of these 1888 varieties tolerant to one or more biotic and/or abiotic stresses, 407 varieties have been developed specially for extreme climate including flood/ water submergence/ water logging tolerance (73), drought/moisture stress/ water stress tolerance (220), salinity/ alkalinity/ sodic soils tolerance (52), heat stress/ high temperature tolerance (49), cold/ frost/ winter chilling tolerance (13) using the precision phenotyping tools. In horticultural crops also, 22 climate resilient varieties have been developed. This includes varieties of potato (5), tomato (2), spinach (1), radish (1), cassava (4), sweet potato (2), greater yam (1), white yam (1), coconut (3) and taro (2).

Further, to tackle the widespread problem of malnutrition in globally, concerted efforts have been made to develop varieties rich in various essentially required micro-nutrients (Protein, Fe, Zn, Vit. A, Vit. C, oleic acid, anthocynin, lysine, tryptophan, removal of anti-nutritional factors like erucic acid, glucosinolates, KTI) for proper growth and development of the body and as a result 124 field crops and horticultural crops varieties have been released and notified during past nine years [Rice (10), Wheat (41), Maize (20), Pearl millet (9), finger millet (4), little millet (1), Lentil (2), Mustard (8), Linseed (1), Soybean (7), groundnut (3) and 18 Horticultural crops]. Concerted efforts for seed production have been made to take the biofortified varieties in general cultivation, which have resulted in coverage of more than 10 million hectares during 2022-23. Biofortified varieties have more than 50% share in total wheat breeder seed indents.

1. Implementation of GAP

The ministry of Agriculture and Farmer's Welfare and Processed Food Products Export Development Authority (APEDA), Govt. of India implements the GAP. APEDA adopts the approach of developing product specific clusters in different agro- climatic zones of the country to help in dealing with various supply side issues through Good Agriculture Practices addressing issues such as, soil nutrients management, higher productivity, adoption of market-oriented varieties etc.